Ghislain THEUBO

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Professional Summary

Former **Dassault Systèmes** R&D intern on **3DEXPERIENCE / 3DSwym** (specs, modular architecture, testing, performance analysis on a major collaborative module rewrite), seeking CDI in **ENOVIA Interoperability R&D (Aix-en-Provence)**. Solid **C++**, algorithms, architecture, and performance analysis from INRIA systems work and personal conversion pipelines; capable of turning robust specifications into reliable converters and APIs. English technical communication. Additional platform SaaS experience at Shift Technology and production backends at Gustave Roussy.

Core Strengths

C++ (OOP, STL, templates) · Algorithms & architecture · Data conversion / interoperability · Specs & documentation · Performance & debugging · JavaScript basics · English · Rigor & autonomy

Experience

Dassault Systèmes – 3DEXPERIENCE / 3DSwym R&D — Vélizy-Villacoublay, France
Software Engineer Intern (Full-Stack Module Rewrite) — *Mar 2025 – Sept 2025*

- Full-stack overhaul of the legacy **Survey module** for **3DSwym** (3DEXPERIENCE collaborative platform): analyze, design, develop, and deliver a modern replacement.
- Backend in **Java 17+ / Spring Boot** (MVC, Spring Data JPA) + **Hibernate**; applied **TDD** with JUnit; design patterns for modular maintainability.
- User-needs analysis with product "Champions", competitive benchmark, MoSCoW prioritization in Agile ceremonies; Git + CI/CD practices.
- Results: delivered core features of the new module with **∼35%** performance improvement and **∼85%** test coverage — robust, maintainable codebase for 3DSwym users.

Shift Technology – Platform Software Engineering — Paris, France
Software Engineer (CDI – Platform / Backend) — *Sept 2026 – Nov 2026*

- Contributed to the **platform / foundations** squad building shared services used by product teams: design and implementation of **C# / .NET** components and **high-performance APIs** on **Microsoft Azure**.
- Worked with cloud data and messaging building blocks (**Azure Cosmos DB**, **Service Bus**), configuration/transversal services, and infrastructure-as-code awareness (**Terraform**) in a hybrid SaaS environment.
- Practiced production engineering habits: design discussions, code reviews, monitoring mindset, incremental delivery, and collaboration with platform and product engineers.

Gustave Roussy & CentraleSupélec MICS Lab — Paris / Saclay, France
Software Engineer (CDD – Backend / Data Platforms) — *Jan 2026 – Sept 2026*

- Software Engineer (CDD) in joint coordination between **Gustave Roussy (Oncostat / CESP)** and **CentraleSupélec MICS**: built and maintained production backends for research/clinical data workflows.
- Designed **REST APIs** (Python/FastAPI) with validation, RBAC, structured logging, and observability; PostgreSQL modeling with indexing and query tuning (**∼30%** faster critical reads).
- Containerized services with **Docker**; automated tests and delivery via **GitHub Actions**; wrote technical documentation for data flows and operational handoff.
- Collaborated daily with domain experts (research/medical) — translating ambiguous needs into reliable, auditable software.

INRIA Grenoble – Humanlab / Froll'N'Roll — Montbonnot-Saint-Martin, France
Research Intern – Real-Time Control Modules (Assistive Wheelchair) — *May 2024 – Aug 2024*

- Developed real-time control modules in **C/FreeRTOS** and Python for an electrically assisted manual wheelchair — direct user impact with continuous validation with the end user.
- Redesigned communication/data pipelines under latency constraints; improved responsiveness by **∼40%**; contributed to open/documented research codebases.
- Human-centered engineering: DIY hardware/software integration, iterative prototyping, and clear technical communication in a multidisciplinary SED team.

LIG – Laboratoire d'Informatique de Grenoble — Grenoble, France
Cloud Infrastructure Intern (VM Migration on OpenStack / Grid'5000) — *Feb 2025 – Mar 2025*

- Automated and studied **VM migration** experiments on OpenStack within Grid'5000; analyzed performance vs availability trade-offs for hosted applications.
- Wrote reproducible scripts and measurement notes under supervision of Alain Tchana — deepened virtualization and distributed cloud skills.

Selected Projects

Multi-Format Engineering Data Converter | C++, CMake, Catch2, JSON/XML GitHub
- C++ toolkit mapping engineering schemas (JSON ↔ XML ↔ CSV) with validation and versioned compatibility — interoperability bridges.
- Streaming parsers and memory profiling for large files; thin JavaScript demo UI calling the CLI/API wrapper.

High-Volume Batch Transform Pipeline | C++, multithreading, benchmarks GitHub
- Batch pipeline with worker pools and back-pressure; documented bottlenecks for volumetry/performance.

C++ Polyhedral Mesh Library | C++, templates, KD-Tree, Catch2 GitHub
- Header-only mesh library with KD-Tree nearest-neighbor and graph shortest paths.

Technical Skills

Languages: C++ (primary push), C, Java, Python, JavaScript (basics), SQL, C# (.NET exposure)
Interop: schema mapping, format conversion, validation, versioning, API bridges
Engineering: modular design, profiling, concurrency, Catch2, Git, CMake, Docker
Cloud: Azure platform services notions (Shift)

Education & Certifications

Grenoble INP - Ensimag Grenoble, France
Engineering Degree – Information Systems / Software Engineering (Double Degree) *2023 – 2025*

École Nationale Supérieure Polytechnique de Yaoundé Yaoundé, Cameroon
Master's Degree in Computer Engineering – Top 12% of class *2019 – 2023*

Certifications: Programming Foundations – TDD (LinkedIn, 2025) · Primerli – Insurance Industry Foundations (2025)

Languages & Extra

English: bilingual / professional · **French:** fluent · **Spanish:** intermediate
Extra: academic tutoring (UGA DAEU maths; 4 years mentoring baccalaureate students in Cameroon) · tennis, football